LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics

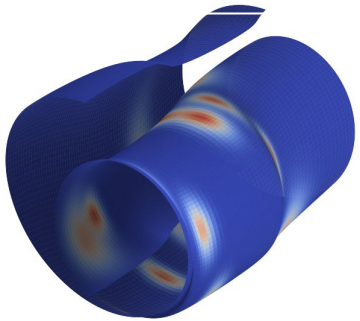
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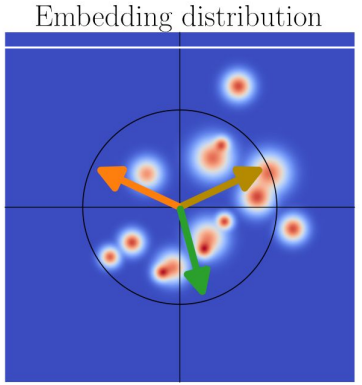
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Learning World Models

- A central challenge in AI is learning manipulable representations of the world and its dynamics from raw observations
- The primary tool for this is mapping high-dimensional observations (e.g., images) to a lower-dimensional embedding space using a deep network, denoted as f_θ.
- The fundamental question is: How should we train the parameters θ without relying on human-provided labels?



$f_\theta \rightarrow$



The JEPA Blueprint (SSL)

- **Joint-Embedding Predictive Architectures (JEPAs)** provide a powerful framework for self-supervised learning (SSL).
- **Core Idea:** Maximize the predictive agreement between the embeddings of semantically related "views" of the data.
- Views can be generated through various transformations:
  - Masking, cropping, blurring (image context).
  - Temporal or spatial translations (video, robotics).
  - Changes in viewpoint or sensor modality.
- The goal is for the learned embedding space to capture the underlying semantic factors of variation in the data.

The Core Problem in JEPAs: Representation Collapse

- The predictive objective in JEPAs has a trivial "shortcut" solution: **representation collapse**.
- The model can minimize prediction error perfectly by mapping all inputs to the same constant vector.
- **Complete Collapse**: All inputs map to a single point.
- **Dimensional Collapse**: All inputs map to a low-dimensional subspace.
- This produces a useless, non-informative representation, defeating the purpose of learning.

Existing Solutions: A Patchwork of Heuristics

- To prevent collapse, existing state-of-the-art JEPAs rely on a complex combination of heuristics:
  - **Stop-Gradient:** Prevents the encoder from directly minimizing the similarity of its own outputs.
  - **Asymmetric Architectures:** Using a "teacher" network (often an exponential moving average of a "student" network) to provide stable prediction targets.
  - **Explicit Regularization:** Whitening or normalization layers to decorrelate features.
  - **Careful Hyperparameter Tuning:** Delicate balance of EMA schedules, learning rates, and regularizer strengths.
- **Result:** Training is often brittle, hard to scale, and lacks theoretical guarantees.

A First-Principles Approach

- Instead of adding more heuristics, this paper questions the fundamental design principles of JEPAs.
- **Central Question:** What are the *necessary conditions* a JEPA must satisfy to learn optimal representations without collapse?
- This approach aims to derive a lean, provable, and scalable JEPA framework from the ground up.

Two Axioms for an Optimal JEPA

- Two minimal conditions are identified as axioms for designing a new JEPA:
  - **Solve the Prediction Task:** The model must still learn to predict embeddings of one view from another, which encourages the capture of semantic information.
  - **Enforce a Target Embedding Distribution:** The embeddings must be constrained to follow a specific, non-collapsible distribution.
- The key is to identify the *optimal* target distribution and a *scalable* way to enforce it.

Finding Optimal Embedding Distribution: Analysis via Linear Probing

- Consider a frozen encoder producing an embedding matrix $\boldsymbol{Z} \in \mathbb{R}^{N \times K}$
- A linear probe is trained to predict unknown targets $\boldsymbol{y} \in \mathbb{R}^{N}$ by solving the Ordinary Least Squares (OLS) problem with Tikhonov regularization:

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^{K}}{\arg\min} \|\boldsymbol{y} - \boldsymbol{Z}\beta\|_2^2 + \lambda \|\beta\|_2^2,$$

- The performance of this estimator depends on its bias and variance, which are functions of the geometry of the embeddings $\boldsymbol{Z}$.

Lemma 1: Anisotropy Amplifies Bias

- **Statement:** For any regularization $\lambda > 0$, whenever the embedding distribution is anisotropic, there exists a downstream task y for which the bias of the estimator $\hat{\beta}$ is higher than if the distribution were isotropic.
- **Intuition:** Anisotropy means the embedding covariance matrix has unequal eigenvalues. Regularization shrinks the parameter estimate $\hat{\beta}$ towards zero. This shrinkage is uniform, but its effect is amplified along the directions of low variance (small eigenvalues) in the embedding space, leading to higher bias for tasks aligned with those directions.

Lemma 2: Anisotropy Amplifies Variance

- **Statement:** With no regularization ($\lambda = 0$), , the total variance of the OLS estimator, $\mathrm{tr}(\mathrm{Var}(\hat{\beta}))$, is minimized when the embedding distribution is isotropic.
- **Intuition:** Anisotropy creates a stretched and compressed embedding space, making the learning process hypersensitive to the specific training data it observes. This instability causes the learned solution to change dramatically with even small variations in the training samples, which results in high variance.

Analysis via Nonlinear Probing

- The analysis extends to more powerful nonlinear probes, such as radius-based k-NN and kernel regression methods.
- The goal remains the same: find the embedding distribution $p(z)$ that minimizes the prediction error, specifically the Integrated Square Bias (ISB), across a family of possible target functions.

Theorem 1: Isotropic Gaussian Optimality

- **Statement:** Among all distributions with a fixed total variance, the isotropic Gaussian is the unique minimizer of the integrated square bias (ISB) for both k-NN and kernel regression probes.
- The integrated square bias is proportional to the Fisher Information of the embedding density, $J(p)$:

$$ISB \propto J(p) = \int \|\nabla \log p(z)\|_2^2 p(z) dz$$

- The isotropic Gaussian distribution uniquely minimizes $J(p)$ for a given covariance constraint.

The Practical Challenge: Enforcing the Target Distribution

- Now that the optimal distribution is identified, how can it be enforced on high-dimensional embeddings during training?
- Standard methods for distribution matching face the curse of dimensionality, becoming computationally intractable or statistically unreliable as the embedding dimension K grows.

Sketching a High-Dimensional Problem

- The key idea is to reduce the multivariate problem to a set of simpler univariate (1D) problems using random projections.
- Cramér-Wold Theorem (Lemma 3): Two multivariate random vectors X, Y have the same distribution if and only if all their 1D linear projections are identically distributed.

$$X \overset{d}{=} Y \iff (u^\top X) \overset{d}{=} (u^\top Y), \quad \forall u \in S^{d-1}$$

- This allows us to match a high-dimensional distribution by enforcing that many of its 1D "sketches" match the corresponding sketches of the target distribution.

Sketched Isotropic Gaussian Regularization (SIGReg)

- **SIGReg** is a loss term that formalizes this sketching idea.
- **Definition:** Given a set of random projection directions $\mathcal{A} = \{a_1, \ldots, a_M\}$, SIGReg measures the average mismatch between the projected embeddings and a 1D standard Gaussian, using a univariate statistical test $\mathcal{T}$

$$\mathrm{SIGReg}_{\mathcal{T}}(\mathcal{A}, \{z_n\}_{n=1}^{N}) = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \mathcal{T}(\{a^\top z_n\}_{n=1}^{N})$$

- Here, $z_n = f_\theta(x_n)$ are the embeddings. The target for each projected distribution $a^\top z_n$ is $\mathcal{N}(0, 1)$

Choosing the Right 1D Test ($\mathcal{T}$)

- The choice of the univariate statistical test $\mathcal{T}$ is critical for stability and scalability. Three families are considered:
  - **Moment-based (e.g., Jarque-Bera):** Match moments like mean, variance, skewness, kurtosis.
  - **CDF-based (e.g., Cramér-von Mises):** Compare the cumulative distribution functions.
  - **Characteristic Function-based (e.g., Epps-Pulley):** Compare the Fourier transforms of the densities.

Flaws of Moment and CDF-based Tests

- Moment-based Tests are Unstable & Insufficient:
  - **Insufficiency (Thm 3):** Matching a finite number of moments does not guarantee that the distributions are identical.
  - **Instability:** Higher-order moment calculations involve high powers of the data (z^k), leading to exploding gradients and unstable training.
- CDF-based Tests are Impractical:
  - They require sorting the data, which is computationally expensive ($O(N \log N)$), non-differentiable, and a bottleneck in distributed settings.

The Epps-Pulley Test: A Stable & Scalable Choice

- The Epps-Pulley test compares the Empirical Characteristic Function (ECF) of the data to the target CF.
  - ECF of data $\{x_i\}$: $\hat{\phi}_X(t) = \frac{1}{N}\sum_{i=1}^{N} e^{itx_i}$
  - Epps-Pulley statistic: $EP = N\int_{-\infty}^{\infty} |\hat{\phi}_X(t) - \phi_Q(t)|^2 w(t) d$

- **Key Advantages:**
  - **Differentiable:** The ECF is a simple average of complex exponentials.
  - **Scalable:** Easily computed in parallel with efficient `all_reduce` operations.
  - **Stable:** The loss, gradients, and curvature are uniformly bounded.

Theorem 4: Stability of the Epps-Pulley Test

- Statement: The Epps-Pulley test, when used as a loss, has uniformly bounded gradients and curvature with respect to its inputs.

$$\left|\frac{\partial EP(a)}{\partial z_i}\right| \leq \frac{4\sigma^2}{N}, \quad \left|\frac{\partial^2 EP(a)}{\partial z_i^2}\right| \leq \frac{C\sqrt{\pi}\sigma^3}{2N}$$

- This theoretical guarantee is crucial for ensuring stable training, especially for very deep networks, without needing gradient clipping or other tricks.

How SIGReg Beats the Curse of Dimensionality

- SIGReg is effective in high dimensions due to two key properties:
  - **Smoothness (Thm 5):** The output distributions of deep networks are naturally smooth. For smooth distributions, constraining a relatively small number of random projections $|\mathcal{A}| = O(K)$ is sufficient to constrain the entire multivariate distribution. The error bound decays as $|\mathcal{A}|^{-2\alpha/(K-1)}$, where α is a smoothness parameter.
  - **SGD Resampling:** During training, the projection directions in $\mathcal{A}$ are resampled at each step. This means the cumulative number of unique directions explored grows linearly with training time, providing dense coverage of the embedding space and ensuring rapid convergence to the target distribution.

The Full LeJEPA Objective

- LeJEPA combines the standard JEPA predictive loss with the SIGReg regularizer.
- **Prediction Loss:** A simple mean squared error loss, encouraging all views to predict the average of the global views. Given embeddings $\{z_{n,v}\}$, the mean of global views is $\mu_n = \frac{1}{V_g}\sum_{v=1}^{V_g} z_{n,v}$.

$$\mathcal{L}_{\text{pred}} = \frac{1}{V}\sum_{v'=1}^{V} \|\mu_n - z_{n,v'}\|_2^2$$

- **Total Loss:** A weighted sum controlled by a single hyperparameter λ.

$$\mathcal{L}_{\text{LeJEPA}} = (1-\lambda)\cdot\mathcal{L}_{\text{pred}} + \lambda\cdot\text{SIGReg}(\{z_{n,v}\})$$

LeJEPA's Lean Implementation

- The final algorithm is remarkably simple and free of common SSL heuristics.
- **Key Features:**
  - No teacher-student architecture.
  - No stop-gradient operation.
  - No explicit predictor network.
  - No batch normalization or whitening layers required in the objective.
  - Implementation requires only ≈50 lines of core PyTorch code.

Stability Across Hyperparameters

(a) (Epps–Pulley) **parameters**

integration	num_slices	config/bstat_n_points		
		5	17	41
$[-1, 1]$	512	71.82	72.13	72.04
	2048	72.88	72.30	72.69
$[-3, 3]$	512	73.95	74.16	74.04
	2048	75.02	74.68	74.77
$[-5, 5]$	512	73.71	74.21	74.15
	2048	74.50	74.80	74.77

(b) Number of local/global views

# global_views (V_g) # views ($V = V_g + V_l$)	1	2	4
4	53.06	72.26	–
6	58.65	73.07	73.68
8	64.46	74.24	73.94
10	68.97	74.06	75.08

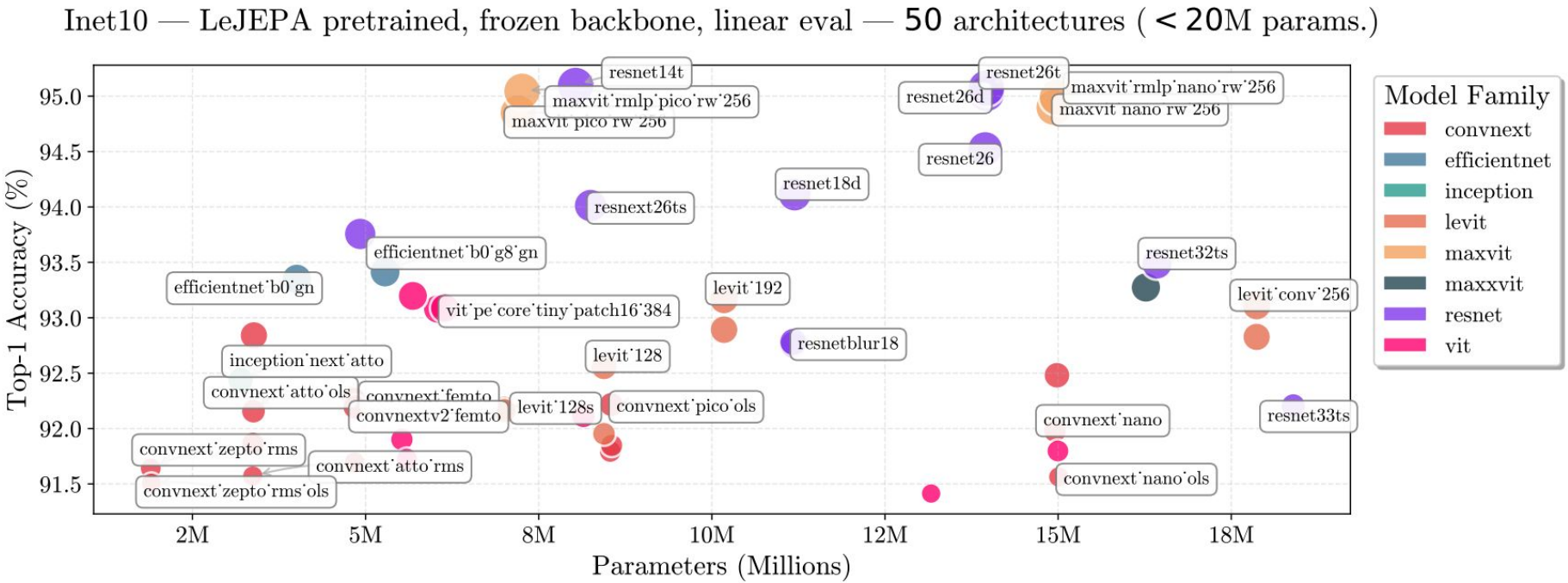
(c) Mini-batch size

batch_size	128	256	512	1024
	72.20	74.15	74.72	74.07

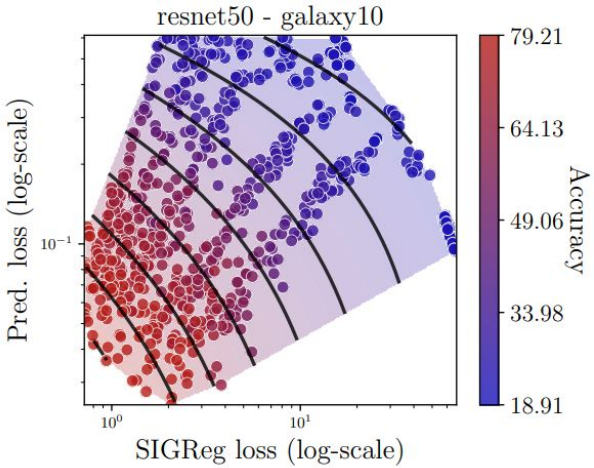
(d) Embedding/Projector dim.

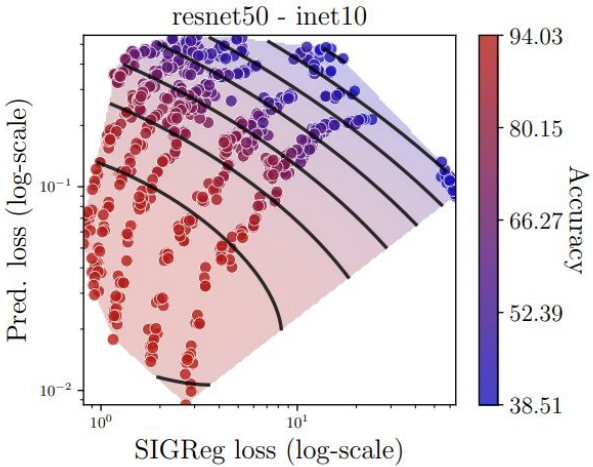
num_slices	1024		4096	
emb. dim. proj. dim.	512	2048	512	2048
64	75.29	75.32	75.50	75.65
128	74.77	75.09	75.26	75.47
256	74.56	74.66	75.08	75.02
512	73.94	74.11	74.81	74.65
1024	73.65	73.94	74.71	74.79

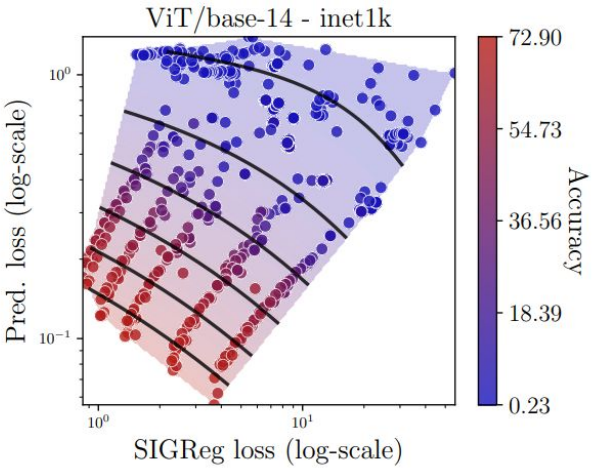
Architecture-Agnostic Performance



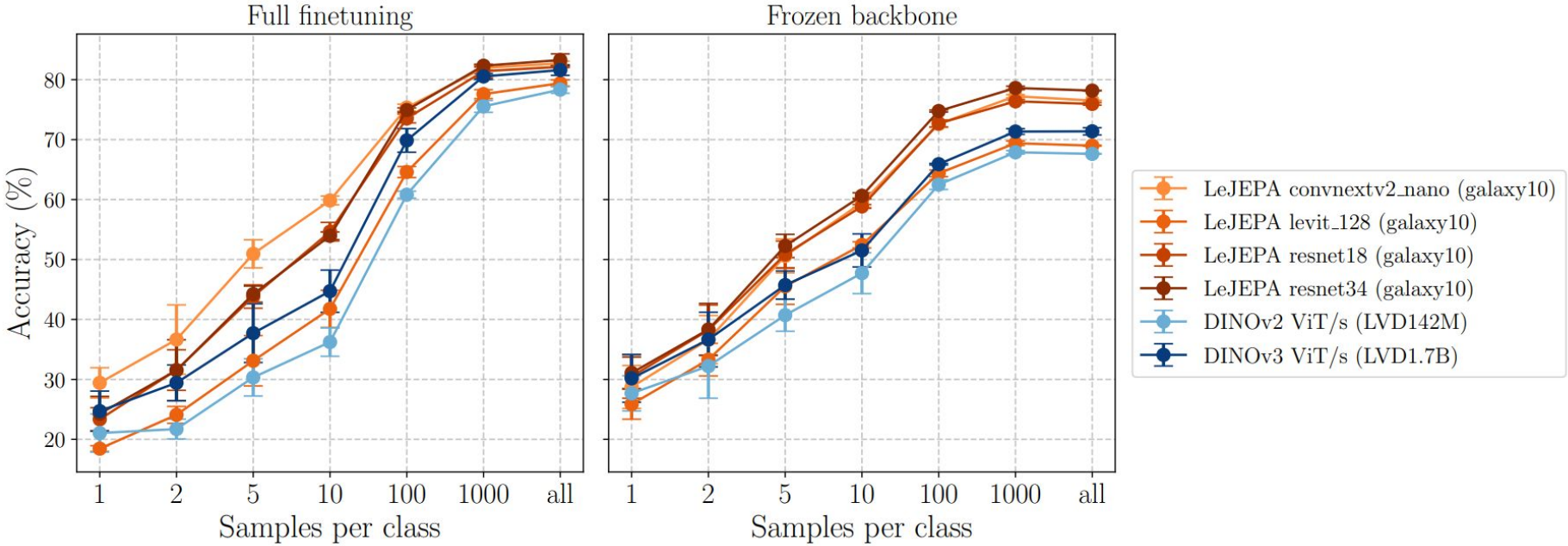
Truly Informative Training Loss







In-Domain Pretraining Outperforms Frontier Models



Competitive Transfer Learning Performance

shots	model	params	pretrain	epochs	Dataset								
					DTD	aircr.	cars	cifar10	cifar100	flowers102	food	pets	avg.
1	LeJEPA ViT-L	304M	IN-1K	100	**33.21**	9.37	3.40	51.65	27.01	48.53	17.14	46.11	29.55
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	32.15	8.07	4.28	50.95	31.48	**48.74**	**17.95**	58.98	31.58
	I-JEPA ViT-H	632M	IN-1K	300	27.71	9.86	4.33	**56.52**	30.58	44.69	14.53	53.38	30.20
	I-JEPA ViT-H + STOP	632M	IN-1K	300	26.60	**11.18**	**4.75**	56.27	**35.20**	47.17	15.75	**59.47**	32.05
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*27.98*	*13.00*	*3.45*	*61.84*	*34.70*	*89.72*	*19.62*	*30.86*	*35.15*
10	LeJEPA ViT-L	304M	IN-1K	100	**64.72**	35.25	22.25	85.15	59.77	**92.53**	**50.90**	77.00	60.95
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	61.84	30.67	24.46	85.74	63.29	91.78	49.32	78.53	60.70
	I-JEPA ViT-H	632M	IN-1K	300	57.68	33.82	21.96	88.77	66.42	88.24	43.97	83.23	60.51
	I-JEPA ViT-H + STOP	632M	IN-1K	300	57.00	**39.77**	**25.21**	**90.09**	**70.32**	90.16	45.68	**85.13**	62.92
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*58.74*	*43.52*	*18.27*	*94.83*	*75.23*	*98.94*	*49.06*	*67.66*	63.28
all	LeJEPA ViT-L	304M	IN-1K	100	**78.30**	57.01	57.28	96.50	83.71	**91.21**	**82.05**	89.74	79.48
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	76.60	52.99	54.88	96.15	81.34	91.11	77.64	89.76	77.56
	I-JEPA ViT-H	632M	IN-1K	300	73.32	56.61	54.47	97.54	86.42	86.47	81.02	92.11	78.50
	I-JEPA ViT-H + STOP	632M	IN-1K	300	73.87	**61.95**	**61.27**	**98.02**	**87.78**	88.08	81.72	**92.88**	80.70
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*75.67*	*65.39*	*49.79*	*98.46*	*89.95*	*98.54*	*81.58*	*87.19*	*80.82*

Emergent Semantic Structure in Representations

